

A Mathematical Framework for CAT & SBI PO

Scoring identities, expected-value rules, sectional attempt plans, and final merit conversion

Abstract

We derive exact scoring equations for CAT and SBI PO, use probability to evaluate question selection, and turn ambitious training goals into explicit attempt/correct/error targets. The scorecards below are hypothetical performance objectives, not predicted percentiles or ranks.

1. CAT: exact score identity

Let A be all attempts, C correct responses, W incorrect penalized multiple-choice responses, and T incorrect non-penalized TITA responses. Correct answers earn +3; wrong MCQs lose 1.

$$A = C + W + T, \quad S = 3C - W$$

Isolate the count of penalized errors:

$$W = A - C - T$$

Substitute and expand the parentheses:

$$S = 3C - (A - C - T)$$

$$S = 3C - A + C + T$$

$$S = 4C - A + T$$

Thus the exact scoring identity is $S = 4C - A + T$. It applies to the sum of all CAT sections, independently of how attempts are distributed.

2. CAT: worked 124-mark target

Metric	Target	Explanation
Attempts (A)	44	Across all three sections
Correct (C)	42	$42 \times 3 = 126$ positive marks
Wrong MCQs (W)	2	2 penalty marks
Wrong TITA (T)	0	Conservative assumption
TOTAL SCORE	124	$126 - 2 = 124$

$$S = 4(42) - 44 + 0 = 124$$

Attempt accuracy = $42/44 = 95.45\%$. If one or both errors are TITA rather than MCQs, the score rises to 125 or 126 respectively. A raw 124 is a goal, not a guarantee of 99th percentile.

3. CAT: sectional training target

Section	Attempts	Correct	Wrong MCQ	Score
VARC	16	15	1	44
DILR	13	13	0	39
Quant	15	14	1	41
TOTAL	44	42	2	124

These are training targets, not quotas to force in the exam. CAT imposes sectional time limits; actual difficulty and institute-specific sectional percentile requirements matter.

4. CAT: expected-value attempt rule

For a penalized MCQ with probability p of being correct, +3 reward and -1 penalty:

$$\mathbb{E}[X] = 3p - (1 - p) = 4p - 1$$

$$\mathbb{E}[X] > 0 \iff p > 0.25$$

For a TITA question without a penalty, expected marks are $3p$. With estimated completion time t , a simplified prioritization score is expected marks per minute. This is NOT classical Kelly optimization: CAT is a finite, timed score-maximization task, not repeated bankroll growth.

$$R_{\text{MCQ}} = \frac{4p - 1}{t}, \quad R_{\text{TITA}} = \frac{3p}{t}$$

5. SBI PO: objective marking derivation

If question group j assigns m -sub- j marks per question, the incorrect-response penalty is one-fourth of its marks. Let C -sub- j and W -sub- j denote correct and wrong counts in that group.

$$S_{\text{obj}} = \sum_j m_j \left(C_j - \frac{W_j}{4} \right)$$

For a single m -mark question, the expected score at correctness probability p is:

$$\mathbb{E}[X] = mp - \frac{m}{4}(1 - p) = \frac{m(5p - 1)}{4}$$

$$\mathbb{E}[X] > 0 \iff p > 0.20$$

A timed heuristic is $m(5p - 1)/(4t)$, where t is expected solving time. Actual Mains questions can have different weights; averages in scorecards are approximations.

6. SBI PO: preliminaries (qualifying only)

Section	Questions	Max marks	Time
English	40	40	20 min
Quantitative Aptitude	30	30	20 min
Reasoning Ability	30	30	20 min
TOTAL	100	100	60 min

Prelims do not count toward final merit. For one-mark questions, $S = C - W/4$. In the referenced 2026 scheme, there is no sectional cutoff in Prelims.

7. Prelims: aggressive sectional hit rates

Section	Attempt	Correct	Wrong	Score
English	36	34	2	33.50
Quant	27	26	1	25.75
Reasoning	27	25	2	24.50
TOTAL	90	85	5	83.75

83.75/100 is an aggressive practice benchmark, not an official cutoff. Target accuracy = $85/90 = 94.44\%$.

8. SBI PO: mains examination structure

Section	Questions	Max marks	Time
Reasoning + Computer	40	60	50 min
Data Analysis & Interpretation	30	60	45 min
General / Economy / Banking	60	60	45 min
English	40	20	40 min
Objective total	170	200	180 min
Descriptive (3 tasks)	3	30	30 min
Mains total	173*	230	210 min

*173 counts 170 objective questions plus three descriptive tasks. The current notification governs eligibility, marking, and section qualification; recheck the next cycle before exam day.

9. Mains: target hit rates (illustrative model)

Section	Attempt	Correct	Wrong	Approx. marks
Reasoning + Computer	33	31	2	45.75
Data Analysis / DI	25	23	2	45.00
Banking Awareness	49	45	4	44.00
English	34	31	3	15.13
Objective total	141	130	11	149.88
Descriptive	3 tasks	—	—	25.00
Mains total	—	—	—	174.88

WARNING: figures above use sectional average marks per question as an illustrative approximation, not actual question-level weights. Actual objective scores depend on which weighted questions are right or wrong.

10. Final SBI PO merit normalization

Let O be Mains objective marks out of 200, D descriptive marks out of 30, and G Phase III group exercise/interview marks out of 50. The weighted final-merit score is:

$$F = 75\left(\frac{O+D}{230}\right) + 25\left(\frac{G}{50}\right)$$

With the illustrative objective plan ($O \approx 149.88$), descriptive target $D = 25$, and Phase III target $G = 40$:

$$F \approx 75\left(\frac{174.88}{230}\right) + 25\left(\frac{40}{50}\right) \approx 77.03$$

This score is a training target, NOT a verified top-100 AIR boundary. Actual rank is determined by the final merit distribution and selection rules.

11. Final screenshot scorecard

Exam stage	Target attempts	Target correct	Score target
CAT	44	42	124 raw
SBI PO Prelims	90	85	83.75 / 100
SBI PO Mains objective	141	130	$\approx 149.88 / 200$
SBI PO Mains descriptive	3 tasks	—	25 / 30
SBI PO Phase III	—	—	40 / 50
Final merit	—	—	$\approx 77.03 / 100$

References: SBI PO 2026–27 detailed recruitment notification (structure/marking/weighting); CAT conventional scoring framework. These score ambitions are not official rank or percentile cutoffs.